VILNIUS GEDIMINAS TECHNICAL UNIVERSITY

FACULTY OF MECHANICS

DEPARTMENT OF BIOMECHANICAL ENGINEERING

STUDY OF THE APPLICATION OF MOTION MEASUREMENTS SYSTEMS FOR GAIT ANALYSIS

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TABLE OF CONTENTS

INTRODUCTION 3
1. DIVISION OF THE WORK AND RESPONSABILITIES 5
2. RELEVANCE OF THE PROBLEM 6
3. METHODOLOGY 12
3.1. Participants 12
3.2. Data Synchronization 13
3.3. Data Processing 14
3.4. Statistical Analysis 15
4. RESULTS 17
4.1. General System Characterization 17
4.2. Evaluation of Spatio-temporal Parameters 17
4.3. Stride Length Analysis 19
4.4. Kinematic Analysis (Joint angles) 20
4.5. Graph Comparative 20
CONCLUSIONS AND DISCUSSION 23
REFERENCES 24

INTRODUCTION

Gait is a fundamental characteristic of human locomotion that serves as a vital indicator of health and functional mobility [6]. Deviations in gait patterns are often associated with neurological, musculoskeletal, or age-related disorders. Consequently, Three-dimensional (3D) motion analysis systems are commonly employed for movement assessments in both clinical and research settings to distinguish pathological conditions from healthy patterns [11].

However, the application of standard gait analysis is often limited by the lack of diverse normative databases. Gait characteristics can be significantly influenced by factors such as race, age, and lifestyle, meaning reference values derived from one population may not be universally applicable. For instance, recent research on healthy Chinese university students highlights that approximately 22.7% report experiencing sports-related injuries annually, yet reference data for this specific young adult demographic remains inconclusive compared to elderly populations [5]. This lack of specific data can lead to overlooked early-stage gait abnormalities and increased injury risks in young adults [5].

To conduct screenings for such specific demographic groups and identify anomalies efficiently, researchers need accessible tools. Traditional Optical Motion Capture (OMC) systems are considered the "gold standard" for precision [1, 2]. However, they are confined to laboratories, expensive, and require time-consuming setups [3]. This makes them unsuitable for widespread field studies or establishing extensive databases for diverse groups like the university students mentioned above.

To address these limitations, Inertial Measurement Units (IMUs) have emerged as a portable alternative [10]. IMUs allow for continuous gait monitoring outside the laboratory, offering great potential for collecting population-specific data in real-world environments [3]. However, IMU systems can be prone to specific errors, such as signal drift or magnetic interference [10]. Therefore, before they can be confidently used to screen populations or diagnose injuries, their agreement with established optical systems must be verified. While previous studies have validated IMUs in specific dynamic scenarios like snowboarding [1] or wheelchair propulsion [2], validation for standard gait across different speeds remains crucial.

Given the increasing utility of wearable sensors, the primary objective of this study is to compare the performance of a full-body inertial motion capture system (Xsens) with a traditional optical infrared system (BTS Bioengineering). Rather than strictly validating one against the other, this research investigates the degree of agreement and kinematic consistency between both technologies in healthy young adults across three different walking speeds (slow, self-selected normal, and fast). Since gait patterns and joint motion characteristics vary significantly with speed [4], assessing the system's stability under these conditions is essential to determine if portable inertial sensors can provide comparable data to laboratory-based systems for future clinical applications.

1. DIVISION OF THE WORK AND RESPONSABILITIES

In the following table, the division of the work and responsibilities is shown.

Table 1. Division of the work and responsibilities

Activity	**Member in charge**
Prepare the participants	Iris Valverde Blanco 60% [redacted] 30% [redacted] 10%
Take measures of the participants	[redacted] 80% Iris Valverde Blanco 20%
Record the participants gaits	[redacted] 80% Iris Valverde Blanco 10% [redacted] 10%
Methodology	[redacted]: 70% Iris Valverde Blanco: 15% [redacted] 15%
Post process data	[redacted] 50% Iris Valverde Blanco 25% [redacted] 25%
Analysis of scientific publication (Introduction & Relevance of the problem)	[redacted] 50% Iris Valverde Blanco 15% [redacted] 35%
Discuss the results (Results and discussion)	Iris Valverde Blanco 50% [redacted] 30% [redacted] 20%

2. RELEVANCE OF THE PROBLEM

Among all the scientific articles that were read, there were two which had more relevance for this work. These were “Comparative analysis of inertial measurement units and markerless video motion capture systems for assessing rotational parameters in snowboard freestyle” by Tom Gorges, Christian Merz, Felix Friedl and Ingo Sandau (2025) and “Validity of IMUs in Comparison to a Marker-Based-Motion Capture System for Spatio-Temporal Parameters During Wheelchair Propulsion” by Lukas Karner Lucas Schreff, Rainer Abel and Roy Müller (2025).

In the first study, “Comparative analysis of inertial measurement units and markerless video motion capture systems for assessing rotational parameters in snowboard freestyle”, addresses the following research question: How valid is a snowboard-mounted IMU compared to a video-based markerless motion capture system (criterion) in measuring the amount of rotation and the course of angular velocity of the board during freestyle tricks on a trampoline? (1).

In order to answer it, eight snowboard freestyle riders were analysed performing air tricks on a freestyle trampoline. Jumps were recorded by eight cameras using Kowa Lenses at 200 Hz and with a factory precalibrated Xsens IMU placed on the bounce board between the bindings using a strap around the board sampling at 100 Hz visualised in Fig. 1.

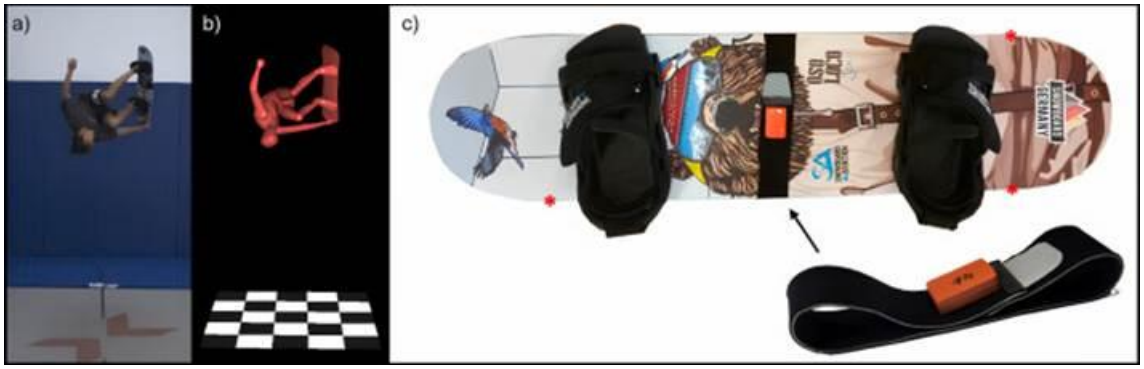


Fig. 1. The snowboard-specific silhouette model and IMU placed on the bounce board

The following parameters were determined:

• Angular velocity motion capture: The angular velocity motion capture (AVmc) is the mean temporal change of the bounce board about the three body axes over the airtime in degrees per second [°/s] calculated from the time series coordinates determined with Simi Shape for every discrete time step (1).

• Angular velocity gyroscope: The angular velocity gyroscope (AVg) is the mean temporal change of the bounce board about the three body axes over the airtime in degrees per second [°/s] calculated from the IMU gyroscope time series data for every discrete time step (1).

• Measured amount of rotation motion capture: The measured amount of rotation motion capture (mARmc) was determined by summing the rotational displacement between the bounce board positions from frame to frame during the duration of the trick (1).

After the data were post-processed with the suitable method for each system and the corresponding calculations were made.

In the second one, “Validity of IMUs in Comparison to a Marker-Based-Motion Capture System for Spatio-Temporal Parameters During Wheelchair Propulsion”, compares a marker-based camera system with inertial measurement units to validate their use in wheelchair propulsion on a test stand (2).

To performance it, twenty-seven participants using their own wheelchair go over distance of 100 m with a consistent movement pattern at three self-selected speeds (slow, normal, fast). Moreover, they were simultaneously recorded using a marker-based camera system and inertial measurement units and subsequently compared between both systems.

A marker-based camera system (VICON, Oxford, UK) was used to monitor the movement of the wheelchair wheels. To assess the rotational dynamics of the wheels, IMUs (Cometa, Bareggio, Italy) were attached to the spokes of both the left and right wheels using fixation plates Fig. 2.

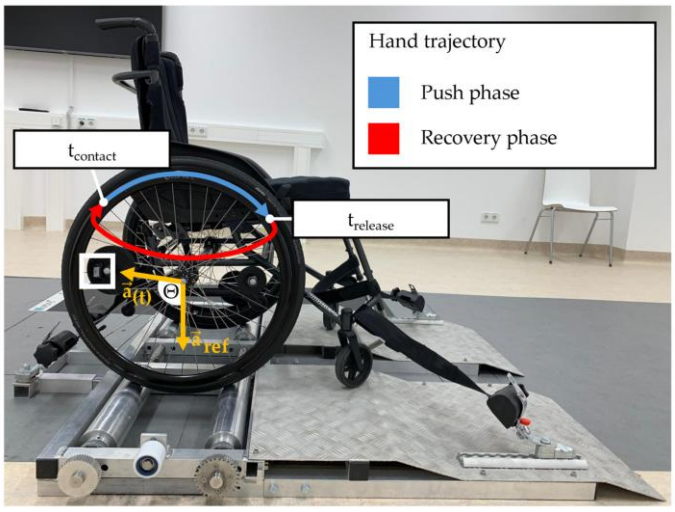


Fig. 2. Wheelchair on a test stand in a laboratory setting, showing markers attached to the spokes and axle and an IMU attached to the spokes, white rectangle. The vectors $a_{(t)}$ and a_{ref} ,marked in orange, are used to determine the angular acceleration based on the calculated angle Θ. (2)

Spatio-temporal parameters like rotational dynamics of the wheels, angular velocity, tangential forces and force exerted by the subjects were determined. After the data were post-processed with the suitable method for each system and the corresponding calculations were made.

Now, the relevance of comparing the two methods has become clear. It is time to design a suitable experiment to compare both measurement systems. Taking as an example the study of Karner, L., Schreff, L., Abel, R., & Müller, R it has been decided to analyse the gait of several participants at different speeds (slow, normal and fast).

Walking is an essential and fundamental activity humans engage in daily activities. However, the biomechanics of walking can vary from person to person, and even small deviations from the norm can lead to gait abnormalities, resulting in pain, injury, and reduced mobility. The ability to walk at different speeds is an important aspect of gait, and biomechanical analysis of gait at different speeds can provide insights into the mechanisms underlying human walking.

Biomechanical gait analysis measures kinematic and kinetic variables such as joint angles, angular velocity, cadence, ground, and muscle activity. Examining these variables provides valuable insights into the fundamental mechanisms of walking. The ability to walk at different speeds is a significant component of gait.

In consequence, it is crucial to comprehend how speed affects the biomechanics of walking. Various speeds necessitate modifications in the kinematic and kinetic variables of walking, such as stride length, stride frequency, and joint angles. Therefore, examining the biomechanics of gait at different speeds is critical for comprehending the impact of speed on walking patterns.

Youth in their twenties walk at different speeds depending on various factors such as distance, urgency, and fatigue. These walking speeds can be categorized as slow, normal, and fast, representing different degrees of exertion and energy expenditure. The effects of walking at different speeds on gait biomechanics in university-aged individuals have not been extensively studied. Therefore, this research aimed to contribute to the knowledge gap by analyzing gait biomechanics in university-aged individuals at different walking speeds using two different methods.

Gait analysis in university-aged students can provide valuable insights into their development and help identify any potential problems in their gait patterns. Many youngsters at this age have suffered injuries in the lower torso, such as broken legs, sprains and ankle dislocations, which can affect their gait. Biomechanical gait analysis can help identify abnormalities in gait development and provide early interventions that can improve walking patterns and prevent musculoskeletal injuries.

An example of the importance of gait analysis is shown in studies of axial spondyloarthritis disease which can lead to spinal mobility restrictions associated with restricted lower limb ranges of motion, thoracic kyphosis, spinopelvic ankylosis, or decrease in muscle strength. The symptoms often start to appear in young adulthood. It is well known that these factors can have consequences on spatiotemporal gait parameters during walking.

On the one hand, one of the main symptoms of axial spondyloarthritis is low back pain. Also, inflammatory back pain is a central criteria for this disease diagnosis. On the other hand, it is also well established that low back pain could significantly affect spatiotemporal gait parameters during walking.

There is evidence of differences in spatiotemporal gait parameters between patients with low back pain and healthy matched controls. Patients with low back pain presented lower gait speed and cadence and shorter stride length than healthy matched controls during walking.

This study patients were total of 30 patients with axial spondyloarthritis and self-reported pain and 30 age and sex-matched healthy individuals. Participants performed 10 meters walk test at comfortable speed.

Various spatiotemporal gait parameters were computed from foot-worn inertial sensors including:

• Speed (ms-1): means walking stride velocity of forward walking.

• Cadence (steps/minute): number of steps in a minute.

• Stride length (m): distance between two consecutive footprints on the ground, from the heel of a foot to the heel of the same foot, one cycle after.

• Swing time (%): portion of the cycle during which the foot is in the air and does not touch the ground.

• Stance time (%): portion of the cycle during which part of the foot touches the ground

• Double support time (%): portion of the cycle where both feet touch the ground.

Data analysis was performed using SPSS (version 20, IBM Corp) and Excel (Microsoft Corp). After the correspondent calculations were made.

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The study “Exploring variations in gait patterns and joint motion characteristics in school-aged children across different walking speeds: a comprehensive motion analysis study” by Mohd Arshad Bari, Haq Nawaz Mir, Junaid Ahmad Parre, Amir Ateeq , Arish Ajhar, Wafa Hashem Al Muslem, Shibili Nuhmani, Anas Alduhishy and Mohammed Essa Alsubaiei shows the relevance of studying the gait at different velocities.

The goal of this study is investigating differences in gait patterns among schoolchildren with different walking speeds and identify the range of motion (ROM) and angular velocity for various joints during gait. To achieve it, a sample of 45 male school-aged children was randomly selected from secondary schools in the Poonch district of Jammu and Kashmir, India (4).

In this work, two FDR-AX700 4K HDR Camcorder video cameras were synchronized to capture the subject's walking gait actions in a defined, specific walkway. The cameras were securely mounted on tripods and fixed to the floor. To ensure the most accurate reconstruction of the two-dimensional coordinates, the cameras were positioned so that their optical axes intersected perpendicularly on both the sagittal and frontal planes (4).

The participants of the study performed a series of walking trials under different conditions: normal (N), slow (S), and fast (F) walking. The individuals were instructed to walk at their typical comfortable walking speed for the normal walking condition. For the slower walking condition, they were instructed to walk at approximately half of their habitual walking speed, and for the fast-walking condition, they were told to walk at their maximum speed. To capture data from a range of walking speeds, participants could choose their walking speeds for slow, normal, and fast walking situations (4).

Thanks to a motion analysis software parameters like joint angles, angular velocity, mean, standard deviation were calculated. In addition, it was investigated the variations between variables and groups, and Post Hoc (LSD) was used to assess significant mean differences.

3. METHODOLOGY

3.1. Participants

Seven healthy adults participated in the study (age 21.3 ± 0.5 years; height 1.71 ± 0.12 m; body mass 68.9 ± 12.0 kg). None of the participants reported neurological, musculoskeletal, or balance disorders.

3.1.1. Experimental Protocol

The aim of this study was to compare gait measurements obtained from two full body motion analysis systems: an optical infrared-camera motion capture system with reflective markers, and a full-body inertial measurement unit (IMU) system. Each participant performed three walking conditions along a 10m walkway: slow walking, self-selected normal walking, and fast walking. For each condition, one valid gait trial was collected, resulting in three recordings per participant. A static calibration trial was recorded at the beginning of each session. Both systems were recorded simultaneously to ensure identical gait cycles.

3.1.2. Optical Motion Capture Systems

Full-body kinematic data were recorded using the BTS GAITLAB system, consisting of a multi-camera infrared optical system (BTS Bioengineering, Milan, Italy) operating at 100–200 Hz. The analysis followed the internationally recognized **Helen Hayes Protocol** (Conventional Gait Model). This protocol uses **18 reflective markers** placed bilaterally on specific anatomical landmarks (e.g., ASIS, greater trochanters, femoral epicondyles, malleoli, calcaneus, and second metatarsal heads).

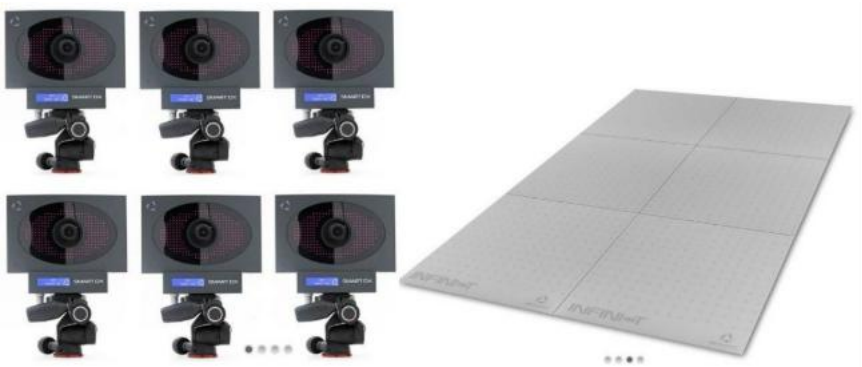


Fig. 3. MoCap System

3.1.3. Full-Body IMU System

A set of 17 wireless IMUs were attached to the head, trunk, pelvis, upper arms, forearms, hands, thighs, shanks, and feet. The sensors sampled at 100–200 Hz. Each participant performed a static N-pose calibration to align sensor and anatomical coordinate systems.



Fig. 4. IMU System

3.2. Data Synchronization

Although both the optical motion capture and IMU systems were recorded simultaneously to ensure identical gait cycles, the data acquisition was initiated manually on the same workstation. This resulted in an unavoidable temporal offset between the start of the two independent recordings.

To correct this latency, a post-processing synchronization method based on kinematic event alignment was applied. The synchronization protocol consisted of the following steps:

1. **Event Identification**: The first prominent gait event (initial heel strike) was identified in both systems. For the optical system, this was determined by the lowest vertical position of the heel marker. For the IMU system, this was identified by the characteristic peak in the vertical acceleration signal corresponding to the impact.
2. **Signal Trimming**: The data streams were cropped to remove the pre-movement static period.
3. **Visual Verification**: Finally, the synchronized signals were visually inspected by superimposing the angular velocity curves to ensure that the subsequent gait cycles remained in phase throughout the trial.

3.3. Data Processing

Data processing was performed using specific software pipelines for each system to ensure clinical validity for the optical system and algorithmic consistency for the IMU system.

3.3.1. Optical Motion Capture Processing

Raw 3D trajectory data exported from the BTS Bioengineering system were processed using a multi-stage pipeline designed to robustly extract temporal-spatial gait parameters from the recorded marker trajectories.

The kinematic data from the optical system was processed using the **BTS SmartClinic** software (BTS Bioengineering, Milan, Italy) following the standard **Helen Hayes Protocol** (Conventional Gait Model). The processing workflow consisted of the following steps:

1. **Static Calibration (Standing):** Prior to processing the dynamic trials, the static acquisition was processed to calculate the **Standing Angles**. This step established the subject's anatomical coordinate systems and segment lengths, providing the necessary offsets to normalize dynamic data.
2. **Trajectory Quality Check (Labeling):** The dynamic walking trials were reviewed using the software's editing tools. Marker trajectories were inspected to verify **correct labeling** throughout the capture volume. Any data gaps (*gaps*) or marker swaps were manually corrected or interpolated using the software's tracking functions to ensure continuous data.
3. **Event Detection and Gait Indices**: The Heel Strike and Toe Off events were indicated to the software for both feet and visually verified to define the gait cycles. Based on these events, the software calculated:
    - Temporal-Spatial Parameters: Stride time, stance phase, swing phase, and cadence.
    - Kinematic Curves: Joint angles for the pelvis, hip, knee, and ankle in the sagittal, frontal, and transverse planes.
    - Gait Indices: The Gait Profile Score (GPS) and Gait Deviation Index (GDI) were calculated to quantify the overall deviation of the subject's gait from the normative database embedded in the Helen Hayes protocol.

3.3.2. Inertial Measurement Unit (IMU) Processing

Data collected from the Xsens MVN Analyze system were processed to extract kinematic parameters using the system's sensor fusion outputs and MATLAB scripting.

Sensor Fusion and Signal Extraction:

The raw IMU data (acceleration and angular velocity) were processed using the proprietary Xsens Kalman Filter (XKF) engine to estimate the robust 3D orientation. Joint angles for the hip, knee, and ankle in the sagittal plane (flexion/extension) were exported directly from the system software.

Event Detection and Parameter extraction:

Gait cycles were identified by analyzing the **knee flexion-extension** signal. The peak knee flexion during the swing phase was detected using the ***findpeaks*** function (MATLAB) with defined minimum peak distance and prominence thresholds to avoid false positives. The stride time is calculated as the time interval between consecutive knee flexion peaks and cadence is derived from the average step time.

Spatial Parameter Estimation:

Due to the drift inherent in integrating acceleration data over long trials, spatial parameters were estimated using a geometric model based on segment kinematics rather than direct integration. The stride length is estimated using a trigonometric approximation based on the hip Range of Motion (ROM) in the sagittal plane and participant leg length and the velocity was calculated dividing the estimated stride length and the calculated stride time.

3.4. Statistical Analysis

Statistical analysis was performed to compare the results of inertial and optical systems.

Descriptive Statistics:

The Mean and Standard Deviation were calculated for all spatio-temporal parameters (velocity, cadence, stride length) and kinematic joint angles across the three walking conditions (slow, normal, fast). For global parameters (velocity and cadence), data from the left and right limbs were averaged to produce a single value per trial. For unilateral parameters (stride length and joint angles), left and right sides were analyzed separately to account for potential asymmetries.

Error Calculation:

To quantify the magnitude of the discrepancy between the two measurement systems, the Absolute Percentage Error was calculated for each variable using the following equation:

$$Error(\%) = \left| \frac{V_{IMU} - V_{Optical}}{(V_{IMU} + V_{Optical})/2} \right| \times 100$$

Variability:

The Coefficient of Variation was computed for stride velocity to assess the consistency of the gait patterns within trials.

All data processing and statistical calculations were performed using custom algorithms developed in MATLAB.

4. RESULTS

The main objective of this project is to assess the performance of a low-cost inertial measurement unit (Xsens) system by comparing it against an optical motion capture system.. In this results section we will be presenting the quantitative results obtained from our measurements and we will also be analysing the behaviour of both systems.

4.1. General System Characterization

Table 2 (below) summarizes the descriptive statistics (mean ± standard deviation) of both systems across all the three different gait speeds.

In general terms, the IMU system demonstrated the ability to replicate the magnitude patterns of the optical system. However, the IMU data has slightly higher variability, more common in the fast gait (larger standard deviations).

Table 2. Descriptive statistics (mean ± SD)

	Slow Gait		Normal Gait		Fast gait	
Variable	**IMUs**	**Optical Camara**	**IMUs**	**Optical Camara**	**IMUs**	**Optical Camara**
Velocity	0.65 ± 0.16	0.67 ± 0.15	1.06 ± 0.45	0.99 ± 0.16	1.30 ± 0.42	1.23 ± 0.38
Cadence	61.92 ± 14.90	66.00 ± 13.74	101.97 ± 27.64	98.40 ± 10.07	116.35 ± 28.05	121.50 ± 24.11
Stride R	1.40 ± 0.38	1.18 ± 0.06	1.30 ± 0.36	1.21 ± 0.06	1.37 ± 0.33	1.18 ± 0.17
Stride L	1.36 ± 0.23	1.22 ± 0.05	1.17 ± 0.25	1.24 ± 0.07	1.32 ± 0.29	1.23 ± 0.11
Hip R	28.85 ± 5.24	27.50 ± 6.95	29.69 ± 7.01	32.14 ± 9.51	34.99 ± 4.13	33.33 ± 7.94
Knee R	40.36 ± 27.83	38.50 ± 28.15	49.98 ± 20.48	47.43 ± 19.60	52.51 ± 24.31	50.00 ± 23.82
Ankle R	10.40 ± 6.30	9.00 ± 5.55	13.05 ± 5.47	9.29 ± 3.45	16.55 ± 6.36	14.17 ± 4.92
Hip L	30.45 ± 8.24	28.83 ± 12.22	31.35 ± 6.31	34.43 ± 9.69	36.58 ± 5.20	35.17 ± 12.77
Knee L	53.88 ± 3.44	56.50 ± 7.48	51.64 ± 18.93	51.00 ± 18.76	53.64 ± 23.68	48.67 ± 20.45
Ankle L	14.61 ± 2.11	11.33 ± 2.94	17.43 ± 3.25	12.86 ± 3.93	19.10 ± 3.11	16.67 ± 5.16

4.2. Evaluation of Spatio-temporal Parameters

Spatio-temporal parameters, specifically mean velocity and cadence, showed the highest degree of concordance between the two systems, meaning that they are the most robust metrics.

4.2.1. Mean velocity

Analyzing the mean velocity, we notice that the IMU system performs more precise estimation at lower speeds but shows a slight tendency to overestimate as the walking speed increases.

As illustrated in Figure 5, during slow gait, the discrepancy between systems is minimal, with the IMUs reporting a mean velocity of 0.65 ± 0.16 m/s, while the optical system reported 0.67 ± 0.15 m/s. In contrast, during fast gait, the inertial system recorded a mean of 1.30 m/s compared to the 1.23 m/s of the optical system.

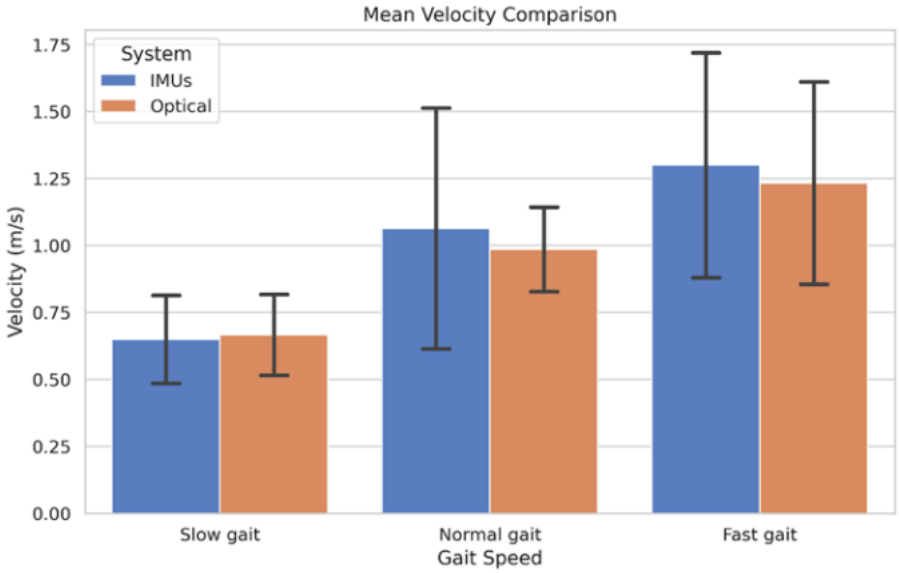


Fig. 5. Comparison of Mean Velocity between IMU and Optical systems across three gait regimes

4.2.2. Cadence

Cadence followed a similar trend to mean velocity. As seen in Figure 6, during normal gait, the inertial system recorded an average of 101,97 steps/min, closely matching the 98,40 steps/min of the optical system. In the slow gait, both system's cadence values also match closely.

However, the error increased noticeably in the fast gait trials. This can be attributed to the higher impact forces and signal noise generated at high speeds, which occasionally caused the peak-detection algorithm to register false positives (extra steps).

Additionally, significant discrepancies observed in some specific subjects were identified as methodological artifacts rather than sensor failures. A mismatch in the data processing window, where the optical system included an extra 'half-step' at the end of the trial that the IMU processing cut off, artificially inflated the calculated error. This highlights the critical importance of precise temporal synchronization and consistent data trimming in validation studies.

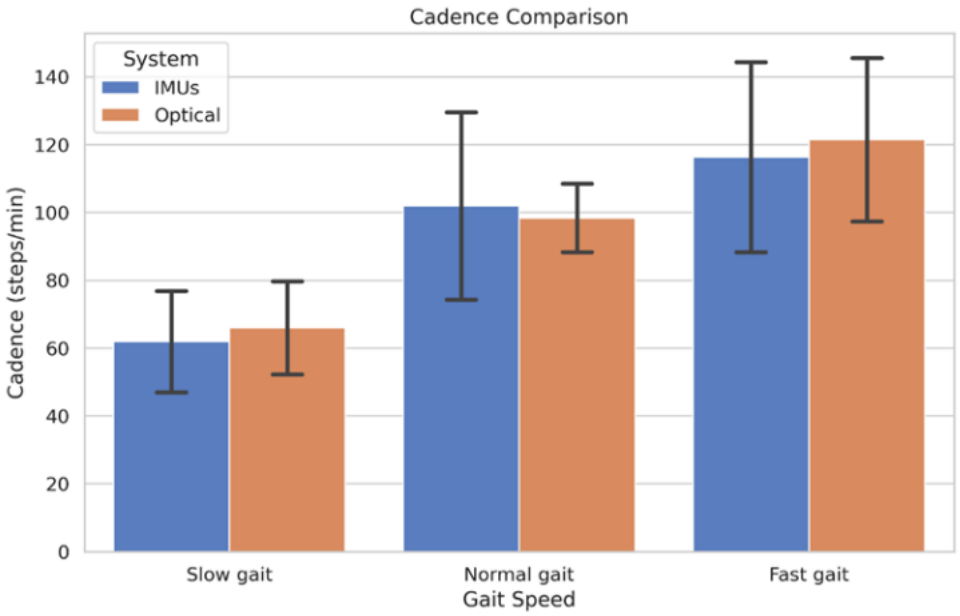


Fig. 6. Comparison of Cadence (steps/min) between systems

4.3. Stride Length Analysis

We evaluated the spatial consistency of the systems following the temporal parameters. As shown in Table 3, the average percentage error for stride length fluctuates between 10% and 20%. The systems maintained symmetry between the left and right limbs regarding stride length, although a systematic difference in magnitude between the two systems was observed, which we will discuss in the following section.

Table 3. Average percentual errors

Percentual Error Mean (%)			
Variable	**Slow Gait**	**Normal Gait**	**Fast Gait**
Velocity	12.78	22.06	7.57
Cadence	7.70	14.45	10.50
Stride R	20.75	19.18	17.96
Stride L	10.78	15.97	20.12
Hip R	22.66	24.12	15.51
Knee R	18.18	7.87	12.18
Ankle R	19.53	30.04	26.46
Hip L	25.20	15.56	21.82
Knee L	7.65	4.90	16.64
Ankle L	26.88	32.30	29.12

4.4. Kinematic Analysis (Joint angles)

The assessment of joint kinematics in the sagittal plane (Flexion-Extension) revealed a clear "proximal-to-distal" gradient in accuracy, as visualized in Figure 7.

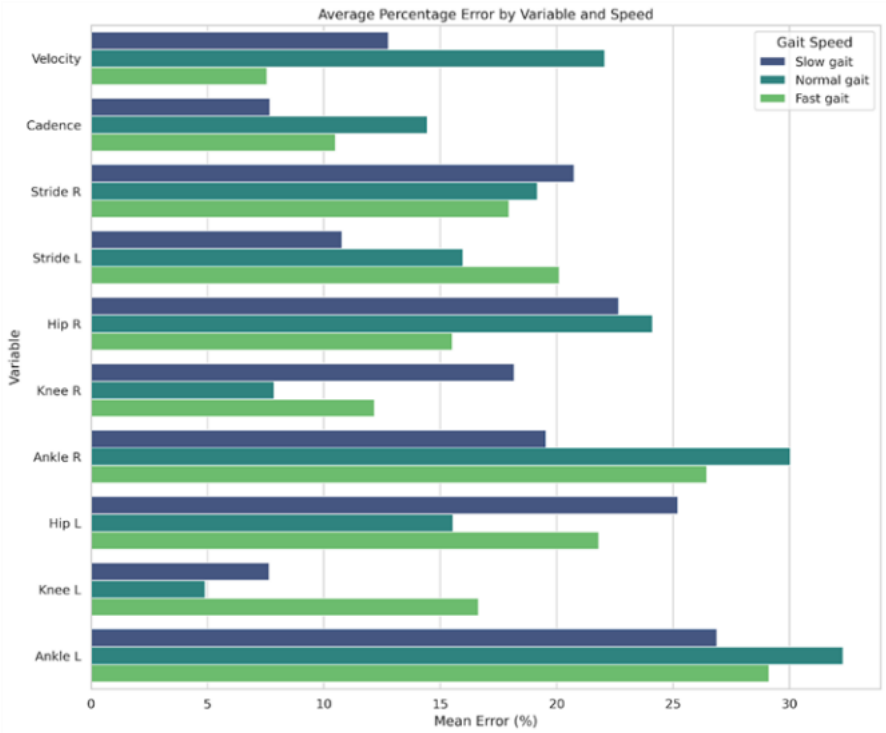


Fig. 7. Average Percentage Error for all variables

The knee joint demonstrated the highest reliability among the angular parameters, particularly at normal speeds. The hip joint showed moderate error rates (approx. 20-25%).

The ankle joint presents the highest average errors, exceeding 30% in the normal gait. As shown in Figure 7, the error bars for the ankle are significantly larger than those for the proximal joints, indicating that the system struggles to reconstruct rapid angular changes in the distal segment.

4.5. Graph Comparative

While the quantitative analysis (tables) provides the magnitude of the error, it does not reflect the temporal quality of the reconstruction. This section presents a qualitative inspection of the kinematic waveforms to validate the post-processing algorithms. The following graphs are not exactly the same between the two systems when comparing them because Motion Capture evaluates by percentage of the cycle and the IMU by time.

Figures 8, 9, and 10 illustrate the knee, ankle, and hip trajectories for subject 2 across the three studied speeds (respectively). Qualitatively, the curves exhibit the expected physiological "peaks and valleys" of a standard gait cycle, such as the double-flexion pattern of the knee.

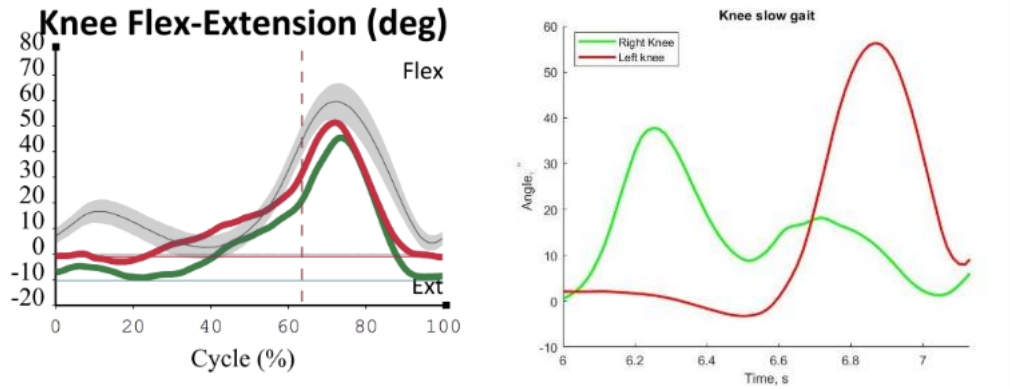


Fig. 8. Knee flexion-extension of participant 2 in slow gait (Optical/IMU)

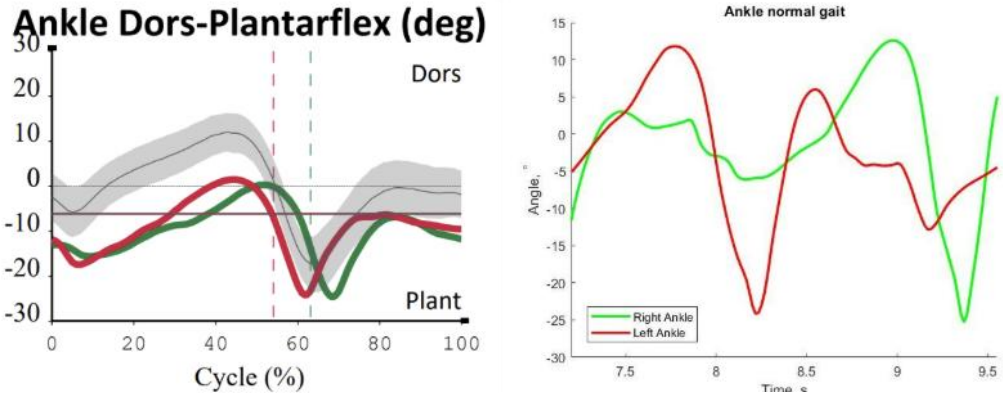


Fig. 9. Ankle Dors-Plantarflex of participant in 2 in normal gait (Optical/IMU)

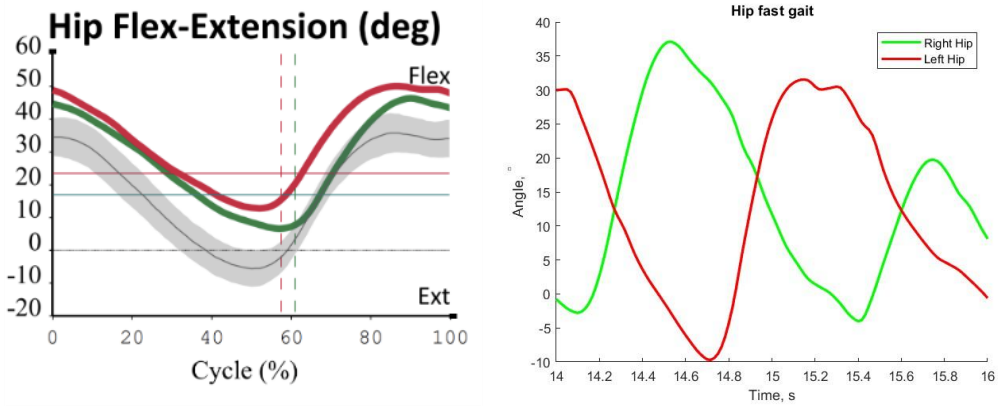


Fig. 10. Hip Flexion-Extension of participant 2 in fast gait (Optical/IMU)

Figures 11 and 12 provide a direct visual comparison between the Optical and IMU systems, respectively, for subject 3's knee flexion. While both systems successfully capture the swing and stance phases, the IMU data shows a slight attenuation in peak amplitude during fast gait. This visual evidence supports the quantitative data: the system correctly reconstructs the movement's "shape," even when numerical errors in peak range occur due to the limitations of the orientation filters.

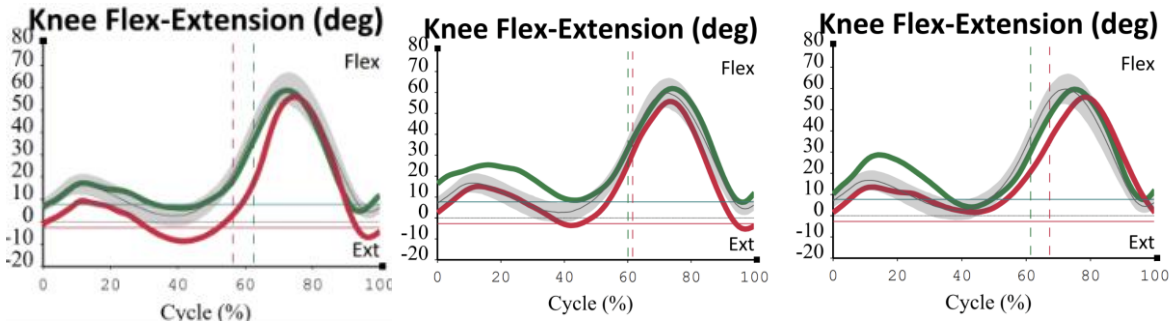


Fig. 11. Knee flexion extension of participant 3 in three different gaits (slow/normal/fast) (Optical)

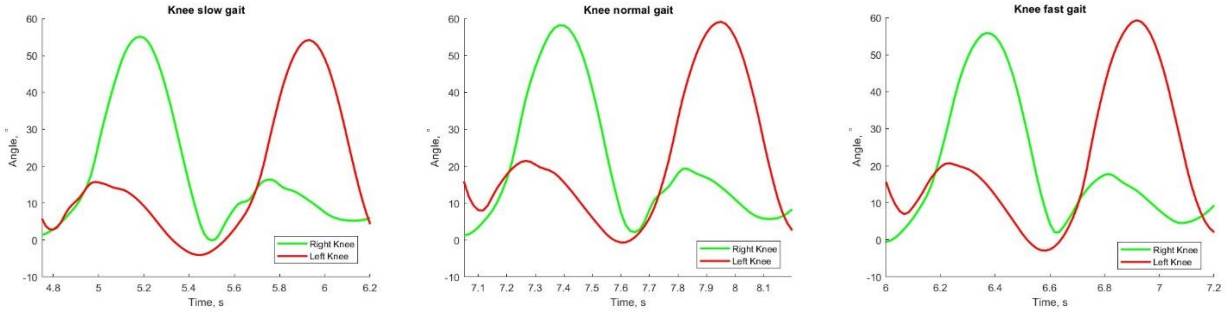


Fig. 12. Knee flexion extension of participant 3 in three different gaits (slow/normal/fast) (IMU)

CONCLUSIONS AND DISCUSSION

The comparative analysis demonstrates that the IMU system is a valid and reliable tool for assessing temporal parameters, such as mean velocity and cadence, during slow and normal walking speeds. Its portability and cost-effectiveness make it suitable for clinical applications, particularly in rehabilitation monitoring, where gait speed is typically reduced and controlled.

Discrepancies in stride length were primarily driven by algorithmic differences: the IMU system utilizes a geometric approximation based on the hip's Range of Motion (ROM), whereas the optical system measures the true Euclidean distance between heel coordinates. Regarding joint kinematics, the ankle consistently presented the highest errors. This distal segment experiences rapid accelerations and higher signal noise during the swing phase, which often exceeds the capacity of current IMU fusion filters. Furthermore, errors in the hip angle were partially attributed to the manual placement of optical markers, as human error in anatomical landmark identification introduces a baseline level of uncertainty into the reference data.

The error analysis in this study was calculated using the mean relative difference, which is highly sensitive to outliers. As observed in our results, specific trials affected by manual synchronization issues or marker displacement produced large error values that skewed the overall mean, potentially suggesting lower performance than reality. Adopting the median as a central tendency metric would provide a more robust estimation of "typical" performance by filtering out these methodological artifacts. Consequently, the reported error rates should be interpreted as a conservative upper bound; errors significantly higher than the average typically represent "methodological noise" rather than inherent sensor flaws.

This study is limited by the small sample size (n=7) and the manual nature of the experimental setup. The manual synchronization of systems and the placement of markers lack the absolute robustness of automated triggers, introducing small temporal misalignments that affect instantaneous value comparisons. Future work should implement automatic synchronization to minimize human-induced error. In conclusion, while the system is not currently recommended for high-precision sports analysis due to limitations at high speeds, it offers a reliable and accessible alternative for general gait assessment in clinical or non-laboratory settings.

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